On the Comprehensive Balance Sheet Stress Testing and Net Interest Income Risk Attribution*

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Abstract

The joint stress testing of net interest income interest rate risk and profit and loss from behavioral risks on a multi-horizon scenario path poses great challenges in enterprise stress testing and earnings risk attributions. We propose a framework for granular level stressed net interest income calculation and P/L risk attribution. The proposed framework can accommodate net interest income impact from interest rate risk, profit and losses from behavioral risks e.g. prepayments and credit defaults as well as facility and deposit contingent drawdowns at each horizon on a given scenario path. Our net interest income framework uses the matched maturity asset and liability concept to synthetically separate the loan origination risks and the Treasury strategic funding risks. It considerably simplifies the joint market and credit risk stress testing by focusing on the asset versus maturity matched liability net interest income on the asset side and the maturity matched asset versus actual liabilities cash flows on the Treasury side. In this framework there is no need for exact cash flow generation on the asset side. The focus is on the loan net interest income process which requires only the outstanding balance. The loan net interest income process is a stochastic process with behavioral events. At the time of the events, the net interest income process regime-switches to capture the event impact e.g., credit loss or prepayment with associated interest rate impact. The event based net interest income process is suitable for loan-level behavioral state path simulation models with competing risk. We also consider the corresponding net interest income process with conditional default and prepayment rate models. The obtained net interest income process can be applied to both loan-level models as well as pools of homogenous loans. We illustrate the framework with competing hazards and state transition models.

*The opinions expressed in the paper are those of the authors and do not necessarily reflect the view of SAS Institute.

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1 Introduction

In this paper we develop a framework that complements comprehensive balance sheet stress testing with a focus on attributing total account level net interest income on a scenario path to financial risks arising from behavioral risks and Treasury funding risks. While regulatory stress testing exercises usually expect aggregated information, the granular level information may either be required for bottom-up approaches or help banks align stress testing results to business profit and loss management. Comprehensive balance sheet stress testing at granular level is complex in practice. In many cases the stress testing of credit risk and prepayment risk in loans has to be handled separately from the banks total interest rate risk stress testing. This approach either ignores the interaction between behavioral risks and interest rate risks, or involves an iterative error prone process.

Regulators, especially central banks, develop their own balance sheet stress testing approaches that focus on the macroeconomic forecasts and the resulting stressed income statement impact. These models however rely mostly on aggregated balance sheet information. See for example, the Bank of England Risk Assessment Model of Systemic Institutions (RAMSI) (2012) and the Dutch Central Bank (De Nederlandsche Bank) top-down stress testing framework (2017). The Bank of International Settlements (2018) reviews the regulatory stress testing practices that include both microprudential and macroprudential stress tests. The regulatory stress testing main components include: Scenario input, risk models, loss projection, net interest income impact and pre-provision net revenue, balance sheet projection and feedback loops e.g., contagion and spill-over effects.

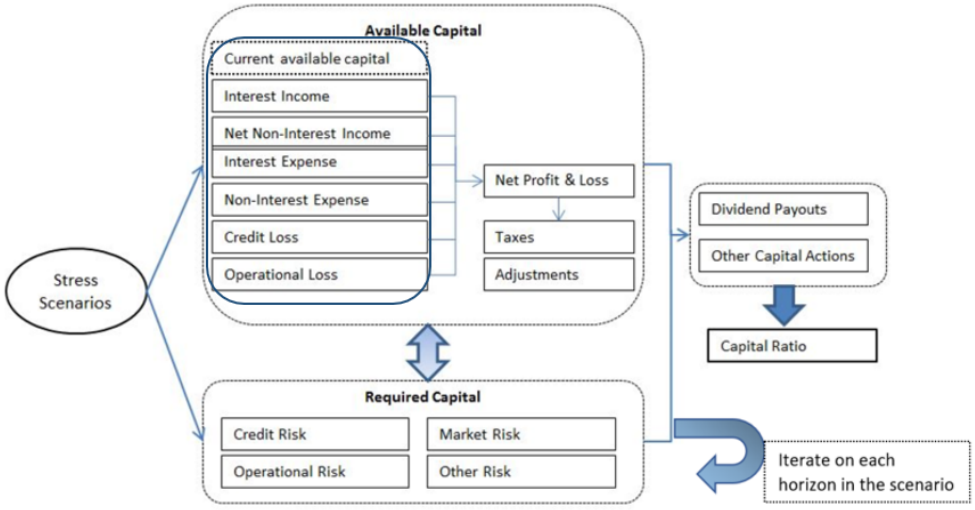
Interest rate risk and behavioral risks including prepayments and credit defaults are the major financial risks that financial institutions must manage at daily basis. However traditionally these risks are managed by Treasury and risk departments in silos neglecting the interactions. The advances in the forward-looking risk management through enterprise stress testing such as the US Comprehensive Capital Analysis and Review[1] (CCAR) and Bank of England[2] stress testing expects aggregation of net interest income, net non-interest income and credit losses on each scenario horizon over a path of the stress testing for the provision and available capital calculation. Figure 1 illustrates the capital adequacy stress testing for each scenario horizon on the path. One primary reason for the complexity is the impact on unpaid balance at each horizon from behavioral risks that affects the interest income in the following stress period, while interest rate changes can also influence the behavioral risks at the same time. Many institutions e.g. the US CCAR banks try to develop an integrated approach to net interest income and behavioral risk analysis or introduce an iterative run-and-adjust approach between the asset and liability management and risk systems. Both approaches exert a heavy burden on the stress testing infrastructure. In this paper we develop a novel scenario-based framework which captures the interest rate and behavioral risk through synthetic separation of asset and liability side of the business. Our approach is founded in classical transfer pricing concepts but extended to include behavioral risks. Funds transfer pricing has a long history in banking for performance measurement and management. See Dev and Rao (2006). Skoglund and Chen (2015) chapter 7 focus on risk based pricing which covers interest rate risk as well

[1] https://www.federalreserve.gov/supervisionreg/ccar.htm

[2] https://www.bankofengland.co.uk/stress-testing

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Figure 1 Capital adequacy stress testing



as behavioral risks. Nyström and Skoglund (2006) apply a matching asset-liability concept specific to credit risk for credit risk economic capital models. By constructing a mirror liability to the asset with the same payment dates and principal payment amounts their focus is on the net interest income margin with default risk.

The proposed framework can accommodate comprehensive net interest income impact from interest rate risk, profit and losses from behavioral risks e.g. prepayments, credit defaults, facility and deposit contingent drawdowns at each horizon on a given scenario path. It simplifies the analysis conceptually and programmatically. The framework can be applied to either account or pool level, and connects well with the existing funding, P/L, provisioning and capital and funds transfer pricing analysis methodologies. Therefore, in addition to enterprise stress testing this framework can be used to attribute total P/L risk of loans or deposits to each risk component for risk adjusted performance analysis and portfolio or product management uses.

Once we attribute to different risk components, we categorize them to different profit and loss effects and manage accordingly. The credit losses are booked against provision and credit capital, and should be built into the loan pricing. In a stress test the loss will affect directly the provision and available capital that carries on to the next horizon. The prepayment, deposit withdrawals and Treasury interest rate mismatch are booked against net interest income, offset by interest rate hedging. In a stress test, it affects the retained earnings and thus impacts available capital in the next period.

Our framework covers two sources of net interest income. The first portion is the traditional lending and deposit business net interest income. The second portion is the Treasury net interest income. An origination asset is linked to a hypothetical matched maturity funding. The hypothetical funding matches the contractual principal cash flows of the asset. Similarly, an origination liability has a matching hypothetical asset. This net interest income construction allows us to separate funding and behavioral risks, yet allow the computation of their combined effect on the net interest income. Another motivation is profit and loss attribution.

Our net interest income process definition considerably simplifies the joint market and

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credit risk stress testing. Firstly, it avoids a joint valuation of banks assets with the bank actual liabilities. This is important as the models used in banks credit impairment forecast and credit stress testing are usually separate from the asset and liability management system that has the bank liabilities. An asset and liability management system is rarely capable to run the complex asset credit models. Instead, the focus is on the funding risk - consistent with our synthetic decomposition of net interest income. By matching the asset with a hypothetical matched maturity liability we can avoid exact cash flow generation in behavioral risks model assessment. This is because of the cancellation of cash flows with the synthetic liability. The focus is on the outstanding balance and the net interest income. Instead, cash flow differences arise between the synthetic asset versus the actual liabilities where the funding risks reside. This is not a concern as cash flow generation and the associated funding interest income analysis are standard features of asset and liability management systems.

The outline of our paper is as follows: In section 2 we describe the net interest income framework for assets, liabilities and the synthetic separation of net interest income. To clearly explain the framework we focus initially on assets with certain contractual cash flow delivery and then deposit liabilities. The assumption of certain contractual cash flows is arguably too simplistic but it helps introduce the concept - especially the separation of net interest income into origination risks and Treasury strategic funding risks. In section 3 we focus on the account-level stressed net interest income with behavioral risks. Behavioral risks include for example competing default and prepayment risks. In our setting a loan does not have to exit completely at a risk event. This is in general consistent with the treatment in forward-looking stress testing requirements that uses the impairment models developed for the international accounting standard IFRS 9 and the US Current Expected Credit Loss (CECL)[3]. In other words, there are no absorbing states. In addition, for a contractual maturity T, if a risk event takes place at $t < T$, the event portion of the loan can be re-originated to the same type of loan, regarding the credit quality as well as risk segment. This treatment is amenable to maintaining balance projection such as constant balance assumption in certain regulatory stress testing. It is also central in our framework as it allows us to keep a balance between the asset and synthetic liability. When a risk event happens at time t the loan-level net interest income changes due to the event. If no behavioral risk event is realized until maturity T the contractual net interest income remains as expected during the entire lifetime. Section 4 applies our stress testing framework to two common types of behavioral risk models used in commercial and retail banking business. In practice a wide range of behavioral risk models are used by banks. Ranging from the more simpler default and prepayment rate models to stochastic simulation models of events. Both can be accommodated in our framework which is illustrated in our numerical example implementations. Finally, section 5 concludes the paper.

2 Net interest income analysis with funding risk

Our stress testing framework is founded in the ability to decompose stress test results into granular level contributions. In practice it is important to understand where the comprehensive net interest income stress test main contributors reside. For that purpose we will use a

[3]The stress testing based on the IFRS 9 models has the additional complexity of stage transfers at the end of the macroeconomic stress test horizon. See Skoglund and Chen (2018).

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synthetic decomposition of the balance sheet into components where assets are synthetically funded by matched maturity liabilities. Similarly, liabilities are backed by matched maturity assets. Our framework helps to understand where in the balance sheet risks arise. For example, if the stress is coming from the lending practice or the Treasury strategy.

Our approach splits the asset net interest income into two sources. Specifically, we consider the following view on the balance sheet

$$(R_A - R_f) + (R_f - R_L) + P/L. \tag{1}$$

Here the balance sheet interest income is coming from income producing assets, R_A. The income producing assets typically have credit risk, prepayment risk and interest rate risk. The term R_f is a synthetic maturity-matched liability. It removes the interest rate risk of the income producing assets with a replicating liability. If the asset continues to produce contractual income during its life, the difference between $R_A - R_f$ is the bank's earned spread in currency amount over R_f which is the funding cost of the replicating liability. This net interest income stream is one of the components of the banks total net interest income. When banks price an income producing asset, its spread must in general cover the expected loss as well as the opportunity cost for the capital held for unexpected loss. This is because if the income producing asset defaults the bank still have to pay the rate on the matched maturity liability until its contractual maturity. This is reflected in the first term of equation (1).

The second term in equation (1), $R_f - R_L$, is the difference between the hypothetical matched maturity asset R_f and the actual liabilities, R_L. Unlike R_f the actual liabilities, R_L, are not tied to a particular loan or asset. This term is the second source of net interest income for the bank. The mismatch between the portfolio of R_f and R_L creates traditional asset and liability management risks on the banks balance sheet. To hedge interest rate risks in the term, $R_f - R_L$, banks enter into trades such as forward rate agreements, futures and swaps. Their impact on the total net interest income of the bank is through mark to market profit and loss i.e., the P/L term in equation (1). A profit and loss effect can also come from the banking book assets that are classified as fair value. See for example Tschirhart et al. (2007), Tilloca (2015) and Skoglund (2017) on the loan fair value practices. Since hedging assets are denominated in market rates rather than the banks funding costs basis risks arise.

One way to view the equation above is that the term, $R_A - R_f$, represents the loan origination net interest income. At origination the loan is funded from the Treasury at the matched maturity rate. Apart from the earnings requirement the pricing of credit and prepayment risks are among the main sources of the spread, $R_A - R_f$. However, there is no funding interest rate risk. The funding interest rate risk is centralized with Treasury. To see this consider originating today a term T fixed rate loan with no credit or prepayment risks such that contractual cash flows at payment times $t = 1, ..., T$ are certain. For an outstanding balance $B_A(t)$ this loan has a net interest income stream at any payment time $1 \leq t \leq T$. The spread income received at time t is:

$$(R_A - R_f) \equiv [(r_f(T) + \beta) - r_f(T)] \Delta t B_A(t-1) \tag{2}$$

where β is the spread rate over the banks funding rate at term T, $r_f(T)$[4] and Δt the time

[4] Although we simply write $r_f(T)$ the funding rate at term T is more specifically a matched maturity funds

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interval between interest payments[5]. The payment at time t is hence made on the outstanding balance $B_A(t-1)$ which represents the balance prior to the time t contractual principal payment and valid during the time Δt. At $t = 1$ we have that $B_A(t-1) = B_A(0)$ the starting balance at origination. By construction the asset and the matched maturity liability have the same payment dates and contractual principal payments. This means that their cash flow difference is solely represented by equation (2). To specify the second term, $R_f - R_L$, we have to specify the actual Treasury liabilities. While the term, $R_A - R_f$, has matching payment dates and principal payment amounts by construction this is not true for, $R_f - R_L$. Hence, general asset and liability management cash flow differences arise. Moreover, the valuation of, $R_f - R_L$, is only relevant at aggregated portfolio level. However, assume for simplicity of exposition that the Treasury funds the single loan in the balance sheet with a matched maturity short-term roll-over liability at payment dates $t = 1, ..., T$, which gives at any time $1 \leq t \leq T$

$$(R_f - R_L) \equiv [r_f(T) - f_A(t-1,t)]\,\Delta t B_A(t-1).$$

Here, $f_A(t-1,t)$, is the forward rate for $\tau > t$ such that $f_A(t,\tau)$ is defined as

$$f_A(t,\tau) = \frac{r_f(\tau)\,\tau - r_f(t)\,t}{\tau - t}.$$

Standing at $t = 0$,

$$f_A(0,1) = r_f(1)$$

while for $t \geq 1$ it is the forward rate. Hence, the floating interest payment at time t is made on the forward rate observed at $t-1$. When interest payments are made in arrears they are instead set on the forward rate, $f_A(t,t+1)$. In this setting the loan net interest income stream, $R_A - R_f$, is locked in at rate β. The interest rate risk is coming from the term, $R_f - R_L$. Specifically, through the realization of the forward rate, $f_A(t-1,t)$. Increasing forward rates reduces the total net interest income. On the other hand decreasing forward rates adds additional interest income in addition to the locked-in spread, β. The Treasury can choose to hedge away funding risk completely by setting $R_L \equiv r_f(T)\,\Delta t B_A(t-1)$ i.e. to fund

rate. For an amortizing fixed rate loan this is in general a weighted average rate such that we could write

$$r_f(T) = r_f(t:T) = \frac{\sum_{t=1}^{T} \Delta B_A(t) \times r_f(t) \times t}{\sum_{t=1}^{T} \Delta_A B(t) \times t}$$

where $\Delta B_A(t) = B_A(t-1) - B_A(t)$ is the principal cash flow at time t, $t = 1, \ldots, T$, and $r_f(t)$ is the bank funding rate at t. The matched maturity funds transfer price is like a strip funding of each principal cash flow weighted by the size and time of the cash flow. In the following it should be clear to the reader when a certain rate is a matched maturity rate or a plain rate.

[5]Without loss of generality we asume all interest rates are expressed as annualized simple rates. We also ignore any discount terms that could be used to present value the net interest income stream. This can be added by multiplying equation (2) with the discount factors

$$d(t) = \frac{1}{(1 + \widetilde{r}(t))^t}$$

with $\widetilde{r}(t)$ the discount interest rate at time t.

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with a matched maturity source. Alternatively, they can keep the short-term funding strategy, $R_L \equiv f_A(t-1,t)\,\Delta t B_A(t-1)$, rolled over time up to T, and hedge the market (interest rate) risk through the P/L term in equation (1) using market instruments. As we have mentioned before when the hedge is through market instruments there is basis risk due to the fact that the banks funding costs are not the same as the market rates. The banks funding costs can be considered as the market rates plus a spread. This funding cost spread is bank-specific and largely related to the bank's own credit quality. Hence, in this situation, changes in the bank funding spread is not hedged.

Loan origination is not the only bank activity. Banks usually try to reduce their market funding cost through low-cost deposit funding. Since a deposit is a liability, we can write the equivalent of equation (1) for a deposit as

$$(R_f - R_D) + (R_L - R_f) + P/L. \tag{3}$$

with R_D the deposit contribution. Essentially, the deposit branch pays the deposit liability rate and earns the bank's funding cost R_f. This is a reward for raising low-cost funding. In a simple idealistic case, consider now a loan branch with the above fixed rate loan with balance $B_A(t)$ and a single term deposit with balance $B_D(t)$. The total net interest income at time $1 \leq t \leq T$ is:

$$(R_A - R_D) \equiv [(r_A - r_f)\,B_A(t-1) + (r_f - r_D)\,B_D(t-1)]\,\Delta t$$

which is the origination earnings on the loan and the term deposit. In a simple idealistic bank setting let $B(t) = B_A(t) = B_D(t)$ for all $t = 1, ..., T$. We then have the net interest income for any t as,

$$(R_A - R_D) \equiv (r_A - r_D)\,B(t-1)\,\Delta t$$

such that the net interest income is due to the origination spread between the loan and the deposit funds. Specifically, we have that

$$(R_A - R_D) \equiv [(r_f(T) + \beta) - r_D(T)]\,\Delta t B(t-1)$$

with $r_D(T)$ the term-T deposit rate. This is an intuitive result in our simple bank framework.

In practice, the bulk of banks deposit liabilities are non-maturity which can be drawn at any time. The withdrawal is partially unfunded and one of the reasons banks need to hold liquid assets as contingency capacity. It is tradition to split the deposit balance into a stable part that is term T asset-backed and a part that is asset-backed short-term. With $B_D(t)$ the stable deposit balance we have the deposit raising net interest income as

$$(R_f - R_D) \equiv [r_f(T) - r_D]\,\Delta t B_D(t-1)$$

with r_D the deposit rate offered for non-maturity deposits. The non-maturity deposit rate is usually not tied directly to the daily market rate fluctuations. It is hence less volatile than market rates. It is a common bank practice to change the non-maturity deposit rate only when there is significant evidence that the market rate level has changed. The assumption of the exact level of stable part of deposits is of course an approximation. The stable part of deposits generally depends on the stressed market situation as well. This is especially true for

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liquidity stress. For the bank, a stable deposit funding base, is a strong contributor to the net interest income. This is due to the generally much lower deposit rates, r_D, than bank funding rates, $r_f(T)$. The non-stable part of the deposit that can likely withdraw at any time i.e., generate an outflow can be viewed as asset-backed short-term. Let $\overline{B}_D(t)$ be the non-stable part of the deposit volume with associated net interest income at time t

$$(R_f - R_D) \equiv [f_A(t-1,t) - r_D]\,\Delta t B_D(t-1).$$

If the non-stable part is withdrawn at time t, the bank must substitute the deposit funding with short-term market funding at rate $f_A(t,t+1)$ at time t. Hence, the P/L impact on the net interest income at time $t+1$ is

$$[r_D - f_A(t,t+1)]\,\overline{B}_D(t)\Delta t$$

such that when deposits are being withdrawn the effect on the balance sheet is to substitute low-cost deposit funding back with regular bank funding. This generally reduces net interest income. The modeling of deposit volumes and the associated net interest income requires models for the three components: market (funding) rates, deposit rates and deposit volumes. See Nyström (2008). Jarrow and van Deventer (1998) considers valuation of non-maturity deposits with the present value of deposits generating value from the difference between the market floating rate and the deposit rate.

We have seen above that when there is no contractual loan cash flow risk (i.e., no behavioral risk) the originated asset-liability loan spread is locked in while the funding risk (interest rate risk) is coming from the Treasury. However, on the asset side contractual cash flows may not be realized as expected due to events like prepayment or default. There is of course embedded interest rate risk in prepayment as the prepayment incentive increases when prevailing market rate lowers. Similarly, interest rate risk is a potential important driver of default rates. We will turn to an expansion of the above simple model to include behavioral risks in the next section. For now, our above synthetic separation of the balance sheet allows us to separate strategic funding risks from behavioral risks. This is important for understanding the contribution of the net interest income stream, and, in particular compute the stressed net interest income at granular asset-liability level.

3 Net interest income analysis with behavioral risks

Our focus in this section is to expand our loan-level net interest income stress testing framework to include risk events such as prepayment and default and the draw-down behavior of the unfunded assets such as facilities.

3.1 Fixed rate loans

Fixed rate loans are in general more complex as they have both prepayment and default risk and is a natural starting point to consider in our analysis. We consider a fixed rate loan with maturity T and yearly payment times $t = 1, ..., T$[6]. The loan is exposed to default risk

[6] The yearly payment interval assumption is made for convenience of exposition and can easily be extended to for example monthly ($\Delta t = \frac{1}{12}$) or even irregular payment intervals. Regardless of payment frequency the synthetic liability always matches the payment dates of the asset.

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and prepayment risk. Specifically, there is a sequence of default times $0 < T_1 < T_2 < \ldots$ and prepayment times $0 < \overline{T}_1 < \overline{T}_2 < \ldots$ that are defined on a complete probability space (Ω, F, P) with right-continuous and complete information filtration $F = (\digamma_t)_{t \geq 0}$. We count the number of default and prepayment times respectively with $n_1 = 1, ..., N_1$ and $n_2 = 1, ..., N_2$. The total number of events is $n = n_1 + n_2$. The nature of the probability measure P depends on the application. In risk management applications, P is the actual or statistical measure. In valuation applications, P is a risk-neutral pricing measure. The financial default loss rate at T_n is given by a random variable $l_n \in \digamma_{t_n}$. Let $\widetilde{T}_1 = \min\left(T_1, \overline{T}_1\right)$ then we can write the realization of the payment period contractual stream of net interest income up until time t as in equation (2) but conditional on the event that $\widetilde{T}_1 > t$. Hence,

$$(R_A - R_f)_{\left\{\widetilde{T}_1 > t\right\}} \equiv \left[\left(r_f(T) + \beta\right) - r_f(T)\right] \Delta t B_A(t-1).$$

Here $r_f(T)$ is the funding rate at maturity T as before, β the loan spread over the funding rate and $B_A(t)$ the loan balance at t. In the following we will suppress the payment interval term Δt due to our yearly payment assumption.

We now introduce prepayment and default risks - starting with prepayment.

3.1.1 Prepayment risks

The early prepayment features can be viewed as embedded options that are American in nature. However, it is not easy to apply embedded option valuation models to non-traded risks such as loans. Also, consumers may not act as rational market players in the sense that the embedded prepayment option is not exercised even when beneficial. This is referred to as customer irrationality and a prime example is that all consumers that should rationally prepay at a certain time don't. Moreover, sometimes consumers prepay when they shouldn't. This phenomena is observed by McConnell and Singh (1994), Stanton (1995), Levin (2001) and, Levin and Davidson (2005). The statistical prepayment models used in practice for loans range from simple fixed portfolio level prepayment rates to loan-specific models capturing the idiosyncratic reasons for prepayment. Such idiosyncratic reasons can include:

- Prepayment (refinance) incentive i.e., current market offered rates vs. loan paying rate.
- Burnout measures i.e., measures of the refinance incentive the loan has been exposed to since the loan was created to understand how sensitive the loans is to the prepayment incentive (customer irrationality).
- Ability to refinance (e.g., current loan to value, credit score, delinquency status (and possibly delinquency history)).

Regardless of the choice of exact prepayment model our key assumption is that the prepayment amount is re-issued at the (matched maturity) forward rate until maturity. At a prepayment event $t - 1 < \overline{T}_{n_2} \leq t$ we hence consider the re-origination at t of prepaid amounts at the forward rate, $f_A(t, T)$, plus the same spread β. For simplicity, we assume the

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spread is fixed over time in our analysis but it can of course easily be assumed variable to capture not only the re-funding risk but also the interest rate reset spread risk[7].

When a prepayment event happens there is no new Treasury funding required. The matched maturity funding rate $r_f(T)$ is still paid on the contractual principal, $B_A(t)$. Hence, if the interest rate decreases (i.e., $r_f(T) > f_A(t,T)$) the prepaid amounts are re-originated at a lower total rate while the locked in matched maturity funding cost remains at the same level. The prepayment incentive is generally increasing with lower interest rates. Hence, in a particular path for the forward rates, $\{f_A(t,T)\}_{t=1}^{T-1}$, that are substantially lower than the initial origination rate with maturity T, $r_f(T)$, we would expect to see an increased prepayment incentive and thus lower anticipated interest income. Once a prepayment has happened, generating interest income at the forward rate $f_A(t,T)$ plus the spread β, we would have to observe further reduced market forward rates to generate a significant prepayment incentive again. With these model assumptions we can write the time $t+1$ loan net interest income at a prepayment event $\overline{T}_{n_2}$,

$$(R_A - R_f)_{\left\{t-1<\overline{T}_{n_2}\leq t\right\}} \equiv \left[\left(f_A(t,T)+\beta\right) - r_f(T)\right] B_A(t) \tag{4}$$

such that a prepayment event switches the forward interest income term, R_A, to the new forward rate plus the loan spread. At each prepayment event the forward interest income term switches and hence the loan net interest income stream can potentially switch at payment times, $t = 2, \ldots, T$. While a prepayment event can practically happen between contractual payment dates $(t-1, t]$ for simplicity we assume the re-origination only happens at t with the prepaid amount. If we assume accrued interest up until t is included in the prepayment amount, then, at time t we still have

$$(R_A - R_f)_{\left\{t-1<\overline{T}_{1}\leq t\right\}} \equiv \left[\left(r_f(T)+\beta\right) - r_f(T)\right] B_A(t-1)$$

conditional on the prepayment event at time t and $\overline{T}_{n_2} = \overline{T}_1$ i.e., the first prepayment event.

Our model can in practice also cater for partial prepayment events by introducing notation for partial prepayment rates that get funded over time with the new forward rate until maturity. We will return to a model of partial prepayments later in the paper.

3.1.2 Default risks

For a default event at T_{n_1} we have that the net interest income at time $t+1$ is:

$$(R_A - R_f)_{\left\{t-1<T_{n_1}\leq t\right\}} \equiv \left[\left(f_A(t,T)+\beta\right) - r_f(T)\right] B_A(t) \tag{5}$$

with the additional loss amount contribution to net interest income entering at time t

$$-l_t B_A(t-1). \tag{6}$$

The loss given default rate term, l_t, in equation (6) realizes at time t while the forward rate impact realizes at time $t+1$ in equation (5). The loss can be an actual instantaneous loss or

[7]Indeed the re-issue of loans at the same spread β is not necessary possible, especially when t gets closer to T (smaller residual maturity) to reflect the diminished behavioral risks and liquidity premium. However, we keep β constant in our exposition for simplicity.

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a booked initial loss that is later adjusted for the realized recovery. Hence, as for prepayment, if this is the first default event and the accrued interest is paid up until t we have the net interest income at t as

$$(R_A - R_f)_{\{t-1<T_1\leq t\}} \equiv [(r_f(T) + \beta) - r_f(T)] B_A(t-1) - l_t B_A(t-1). \tag{7}$$

If there is also loss of payment period accrued interest at t it is added to the loss term such that we have at time t for a default event T_{n_1}

$$(R_A - R_f)_{\left\{t-1<T_{n_1}\leq t\right\}} \equiv -r_f(T) B_A(t-1) - l_t B_A(t-1).$$

Similarly, accrued interest from longer delinquency spells before the default event can impact the total loss and hence the net interest income stream prior to the loss event. When a default event or delinquency happens the synthetic liability and actual Treasury liability still have to be paid. Hence, there is no effect on the Treasury funding, $R_f - R_L$, net interest income. In principle the spread, β, should protect the bank from defaults on portfolio basis. This is the first protection level i.e., to price the credit risk. The second level of protection is capital i.e., a shortfall of principal between $R_A - R_f$ ultimately must be covered. Firstly, the bank takes a hit on earnings and then secondly on the capital base. In equation (5) the assumption is that the loan is re-originated at the current balance at the next payment time following a default event or the payment time at the default event time. Hence, if a default event happens between $(t-1, t]$ there is a loss amount at time t equal to $l_t B_A(t-1)$ as in equation (6). The loss is taken from provisions, or in a worse case from capital, and put into business again. The loan is then re-originated at the then prevailing rate implied by the forward rate. This can be seen from the net interest income equation (5) at $t+1$ following a default event at $t-1 < T_{n_1} \leq t$. Hence, there is no new matched maturity funds. Just cash ingestion to re-originate the loan at the forward rate with balance $B_A(t)$ and pay the time t contractual principal on the funding, $\Delta B_A(t)$. If no new default event happens between $(t, T]$ the net interest income stream for $\bar{t} \in [t+1, T]$ is locked in as

$$(R_A - R_f)_{\left\{\max\left(T_{n_1}\right)\leq t\right\}} \equiv \left[\left(f_A(t,T) + \beta\right) - r_f(T)\right] B_A\left(\bar{t} - 1\right). \tag{8}$$

Apart from the booked financial loss portion due to default at time t this is the same net interest income situation as in the prepayment event. The assumption that credit losses hit the provisions or capital to keep a constant balance balance sheet ongoing respects business continuity. This also keeps the balance between the asset and the hypothetical matched maturity liability which is central. Default risk, even with full instantaneous recovery, is not risk-free. The re-originated fully recovered balance has interest rate risk. This is again consistent with bank business continuity.

The reader may be concerned about potential multiple default effects. However, there is no multiple default effect on the original loan nor the re-originations. Following a default event the new re-origination loan can of course default. If a default happen a new loan is always re-originated. To clearly explain the implications of our re-origination assumption let the loan exit at the first default event between $(t-1, t]$. We then have that the net interest income contribution at time t is (with no loss of period accrued interest) given by equation (7), and, for $\bar{t} \in [t+1, T]$

$$(R_A - R_f)_{\{t-1<T_1\leq t\}} \equiv -r_f(T) B_A\left(\bar{t} - 1\right).$$

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This is a potential way to view the ex-post net interest income performance of the original loan. Specifically, if we assume the default can happen only at the payment times $t = 1, ..., T$, the cumulative net interest income, $(R_A - R_f)_\Sigma$, for the original loan is then

$$(R_A - R_f)_\Sigma \equiv \sum_{j=1}^{\min\{T_1,T\}} \{[(r_f(T) + \beta) - r_f(T)] B_A(j-1)\} - \chi_{t\{T_1=t\}} l_t B_A(t-1) \quad (9)$$
$$-\chi_{t\{T_1<T\}} \sum_{i=T_1+1}^{T} r_f(T) B_A(i-1).$$

with $\chi_{t\{\cdot\}}$ the indicator function. However, equation (9) implies that cash from recovery and provisions are not re-invested and are earning no interest. The cash is just used to match the principal payments of the matched maturity liability until maturity time, T. This means that the bank is accepting a negative net interest income stream at the default event and every subsequent time until maturity. Clearly, the bank is better off with re-investing. If they immediately, at $T_1 = t$, re-invest in the Interbank market in a close to risk-free matched maturity asset they can receive for $\bar{t} \in [t+1, T]$ the net interest income stream

$$(R_A - R_f) \equiv [f_A(t,T) - r_f(T)] B_A\left(\bar{t} - 1\right). \quad (10)$$

This re-investment is done at the prevailing forward rate, $f_A(t,T)$, i.e., the loan is replaced with a (matched maturity) fixed rate risk-free loan with the same residual maturity as the synthetic liability. This adjustment better represents the original loan ex-post net interest income performance. Ignoring interest rate risk for the moment (i.e., $f_A(t,T) = r_f(T)$) we would have

$$(R_A - R_f)_\Sigma \equiv \sum_{j=1}^{\min\{T_1,T\}} \beta B_A(j-1) - \chi_{t\{T_1=t\}} l_t B_A(t-1) \quad (11)$$

with exit at the first default time, T_1. Apart from a discounting term this net interest income equation is used by Nyström and Skoglund (2006) in credit risk net interest income modeling.

Clearly, equation (10) is very close to our assumption of re-investment in a loan with the same credit risk characteristics at a default event in equation (8). The only difference is the additional earned spread β which should be set at least to compensate for the behavioral risks. This is consistent with the risk based funds transfer pricing framework. It also illustrates that our assumption of re-investment is reasonable from a business standpoint. The assumption of re-investment to the same loan keeps the credit risk characteristics of the portfolio constant but is not central to our analysis. The modeler can decide which exact loan is re-invested to. For example, in a stressed market situation the credit risk of loans will generally increase. The bank cannot however stop doing business. If the re-investment to a certain defaulted loan type at stressed times is not good bank business the modeler can replace the loan re-investment to the same loan with another loan that is acceptable. This does not change the fundamental aspects of our framework and the computation of the net interest income stream.

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3.1.3 Competing default and prepayment risk

With competing risk of default and prepayment we can write the net interest income at $t+1$ following an event at $(t-1,t]$ as

$$(R_A - R_f)_{\left\{t-1<\widetilde{T}_n\leq t\right\}} \equiv \left[\left(f_A\left(t,T\right)+\beta\right) - r_f\left(T\right)\right] B_A\left(t\right) \tag{12}$$

with

$$\widetilde{T}_n = \min\left(t-1 < T_{n_1} \leq t, t-1 < \overline{T}_{n_2} \leq t\right)$$

the next default or prepayment event - whichever comes first during the interval $(t-1,t]$, and,

$$-l_t B_A\left(t-1\right)$$

the loss amount contribution at t - conditional on a default event i.e., $\widetilde{T}_n = T_{n_1}$. As before, accrued interest losses are added. In a traditional competing risk framework the effects of prepayment and default on overall attrition are negatively correlated. When interest rate falls more loans prepay but fewer default. Similarly, when interest rates rise prepayment slows but the defaults increase. See Hall and Lundstedt (2005).

We emphasize that for either a default or prepayment event at $(t-1,t]$ the term $\left(f_A\left(t,T\right)+\beta\right)$ in equation (12) represents the re-origination at t at the forward rate, $f_A\left(t,T\right)$, with spread β. In practice, for fixed rate loans, the forward rate can be viewed as a matched maturity forward rate. As we have mentioned before, in our framework, default risk is treated as prepayment with an additional loss event at amount, $l_t B_A\left(t-1\right)$, at the next possible payment t to the default time, T_{n_1}. The cumulative loss amounts up until time t is a point process,

$$\overline{L}_t = \sum_{n_1 \geq 1} l_{n_1} B_A\left(n_1 - 1\right) \chi_{\{T_{n_1}\leq t\}}.$$

When a default loss event happens at time $(t-1,t]$ the loss is booked at time t at loss given default rate l_t. If no new default or prepayment event happens between $(t,T]$ the net interest income stream for $\overline{t} \in [t+1,T]$ is locked in as,

$$(R_A - R_f)_{\left\{\max\left(\widetilde{T}_n\right)\leq t\right\}} \equiv \left[\left(f_A\left(t,T\right)+\beta\right) r_f\left(T\right)\right] B_A\left(\overline{t}-1\right)$$

Hence, it is locked in at the last event forward rate at t.

We finally note that the rundown of loan balance is only due to the contractual principal payments which are usually deterministic. This makes it easy to introduce new loan business to compensate for the principal rundown up to a target balance. In particular, there is no need to run the current book of loans first to observe the actual rundown before new business can be introduced.

3.2 Floating rate loans

While fixed rate loans can have a significant risk of prepayment, and hence interest rate risk in the spread $R_A - R_f$, this is not the case for floating rate loans. The matched maturity funding rate for a floating rate loan is the forward rate, $f_A\left(t,t+1\right)$. Write therefore the net

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interest income equation at $t+1$, as in the prepayment net interest income equation (4) for the fixed rate loan, but now with a floating rate loan funded at forward rates,

$$(R_A - R_f)_{\left\{t-1<\overline{T}_{n_2}\leq t\right\}} \equiv \left[\left(f_A\left(t,t+1\right)+\beta\right)\right] B_A\left(t\right) - f_A\left(t,t+1\right) B_A\left(t\right)$$

which shows that the prepayment has no effect on the net interest income term, $R_A - R_f$, as the loan is already funded with a re-priced matched maturity liability. Hence, for a floating rate loan we have that the conditional default time t net interest income is (with no loss of accrued interest at t)

$$(R_A - R_f)_{\left\{t-1<T_{n_1}\leq t\right\}} \equiv \beta B_A\left(t-1\right) - l_t B_A\left(t-1\right) \tag{13}$$

such that there is no interest rate risk - only plain default loss risk[8]. When there is no default event between $(t-1,t]$ the loan net interest income is simply the margin rate, β, applied on the outstanding balance. Note here that in our model a default event by itself does not necessarily lead to loss if $l_t = 0$ and there is no loss of accrued interest. The default event itself has interest rate risk only in the fixed rate loan case. Effectively, a default event with no loss is like a prepayment effect. The prepayment effect only impacts the net interest income stream of fixed rate loans. If a credit loss happens between $(t-1,t]$ it reduces the time t net interest income. But it is booked as a loss against the provision, not as a reduction of the net interest income. This is true for both fixed and floating rate loans. See Figure 1 on the separation of the net profit and loss into net interest income and credit loss.

3.3 Example of loan net interest income with competing risk events

Our loan net interest income model can be thought of as a regime-switching model where there are certain states that can be occupied at a certain time t when payments are due e.g., contractual payment (current), default event or prepay event. Both the default and prepayment states are temporary due to the immediate re-origination assumption. However, they affect the net interest income realized. Hence, unlike traditional credit risk and prepayment modelling the states are not absorbing. Once an event has happened it specifically changes the net interest income equation until the next competing risk event happens. In addition, a credit risk event also books a loss at the payment time t. After a credit event at $(t-1,t]$ and assuming no new credit event happens between $(t,t+1]$, the net interest income equation at $t+1$ has changed due to the earnings rate switch to the forward rate at t plus the spread i.e.,

$$R_A \equiv \left(f_A\left(t,t+1\right)+\beta\right) B_A\left(t\right).$$

Not only default and prepayment states can be handled in this loan-level net interest income framework but also other competing states such as loan modification and delinquency

[8] While there is no interest rate risk impact from prepayment in floating rate loans there can still be other interest rate impacts that need to be taken into account. For example, floating rate loans sometimes have a cap on the paying rate that needs to the priced into the spread rate, β, in the form of an option adjusted spread. Hence, ceteris paribus, a floating rate loan with a cap should pay a higher loan interest rate than the corresponding plain floating rate loan. The cap also affects stressed net interest income when the forward rate plus the spread rate, β, exceeds the contractual cap rate. Hence, a cap can by itself result in a negative stream of net interest income when funding forward rates are rising and cannot be fully passed on to the loan paying rate.

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with associated accrual of interest. A modified loan usually takes a reduced balance (with part written-off as a loss) and/or at a different rate. This falls into the same framework as our treatment of default risk. Increasingly severe delinquency states means accrual of the contractual interest income portion, R_A, from previous periods. If it is eventually paid back at t it contributes to the current net interest income portion at t by adding the missed payments. If its eventually lost in a subsequent default it contributes to the total loss term.

Figure 2 gives an illustration of a particular loan path that can happen in a competing risk model of credit and prepayment events. In this model a loan can pay as usual (Cur), be delinquent 30 days (D30), be delinquent 60 days (D60) or be delinquent 90 days or more (D90p). The loan can prepay (Prepay) from the payment state (Cur) as well as any of the delinquent states. Once a loan becomes delinquent 30 days or more it can be modified (Mod). A default event can only happen when the loan has been delinquent 90 days or more (D90p). Hence, for a default event to happen a loan must first pass the increasingly severe longer delinquency periods. When a default event happens from the D90p delinquent state there is the calculation of the loss given default. Our model example in Figure 2 focuses on events and is consistent with our loan-level net interest income model. The model representation captures general stochastic models where each state transition probability is stochastic and can in general depend not only on idiosyncratic loan factors and the economic situation but also the path up to the state i.e., the realized history. Such stochastic event simulation models are often solved by simulation due to the path-dependency. When a risk event, and effectively a re-origination, happens with state path history tracking the modeler must decide if the new origination is similar to the current loan issuer history status up to that point or if a fresh start is preferable. A bank can essentially retain the same business mix and portfolio quality by re-originating to the same loan issuer with modified behavioral history close to the credit risk or prepayment event. We will see an example of this in the application section.

Consider now the particular fixed rate loan path illustrated in Figure 2 with the dashed lines to the new occupying states. The time line indicates a path where the loan moves from being current at origination to being increasingly severe delinquent in $s = 1$, $s = 2$ and $s = 3$. At $s = 4$ it defaults and is re-originated at the current unpaid balance. We can now simply compute the net interest income realized from the path in Figure 2 with payment times $t = 1, \ldots, T$ and $t + 1 \geq s > t$. Specifically, for the first payment period $t = 1$ we have that

$$(R_A - R_f) \equiv (r_f(T) + \beta) B_A(0) - r_f(T) B_A(0)$$

and for the subsequent payment periods $t = 2$, 3 and 4 we have

$$(R_A - R_f) \equiv -r_f(T) B_A(t-1)$$

due to the delinquency. There is no accrual of delinquency periods contractual principal payments but it can be added to the final loss term. Assuming that the loss given default rate of the principal is 0.5 we have for $t = 5$

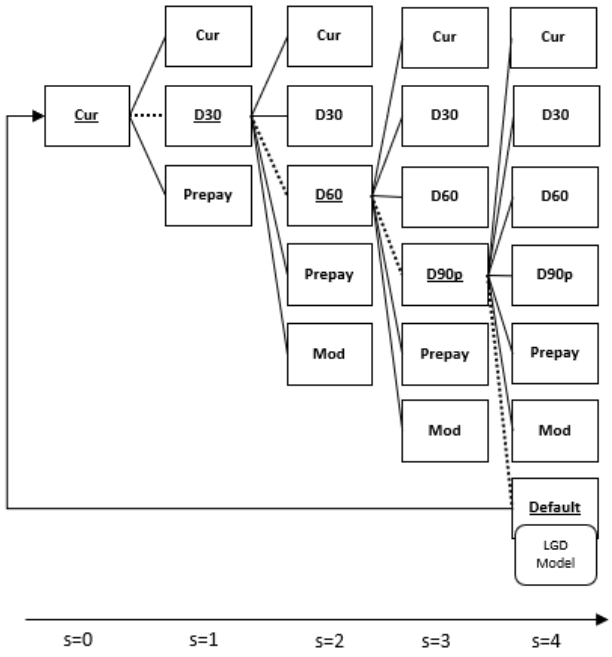
$$(R_A - R_f) \equiv -0.5 B_A(4) - r_f(T) B_A(4).$$

At $t = 5$ the re-origination happens such that if the re-originated loans stays current we have at $t = 6$

$$(R_A - R_f) \equiv (f_A(5, T) + \beta) B_A(5) - r_f(T) B_A(5).$$

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Figure 2 Sample fixed rate loan event path



For simplicity, let the there be a constant balance, $B_A(t)$, of 1 unit of currency and write the $t = 1, ..., 6$ cumulative net interest income as

$$(R_A - R_f)_\Sigma \equiv f_A(5,T) + 2\beta - 5r_f(T) - 0.5.$$

We can now allocate the $t = 1, \ldots, 6$ path cumulative net interest income P/L loss, $\text{NII}_{P/L}$, to credit risk. The cumulative $\text{NII}_{P/L}$ loss is the difference between the promised contractual cash flows and the realized cash flows such that,

$$\text{NII}_{P/L} = 6\beta - (R_A - R_f)_\Sigma .$$

Consistent with equation (11), and allowing for loss of accrued interest up until default, the credit risk P/L attribution to the original loan is

$$\text{NII}_{P/L} = 4\beta - 4r_f(T) - 0.5.$$

Of course, we can allocate the $\text{NII}_{P/L}$ to other events as well as they happen. For example, if the loan stays current for $s = 1, ..., 3$ and then prepays at $s = 4$ the cumulative $\text{NII}_{P/L}$ allocation is

$$\text{NII}_{P/L} = f_A(5,T) - r_f(T)$$

which is the $t = 6$ difference between the re-originated forward rate and the matched maturity funding rate.

3.4 Treatment of unfunded assets

Not all assets are fully funded. Undrawn portions of a facility or credit card need to be funded at the time of the usage. As draws happen they are funded short-term using the matched maturity forward rate, $f_A(t, t+1)$. The additional drawn balance of a facility therefore behaves

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like a float rate loan but at the current drawn balance. Let the additional drawn balance be $B_U(t)$ then we have at time t (with no loss of accrued interest) conditional on a default event

$$(R_A - R_f)_{\{t-1<T_{n_1}\leq t\}} \equiv \beta B_U(t-1) - l_t B_U(t-1) \tag{14}$$

Because the undrawn portion of the facility is funded with short-term rollover at the forward rate there is no prepayment impact. When there is undrawn portions of a facility it is natural to let the undrawn portion depend on the credit quality of the borrower. That is let $B_U(t)$ depend on the state of the borrower. For example in Figure 2 we could see increased usage of the undrawn amount as increased delinquency severity and ultimately default happens. More usage of the borrower facility means in general greater net interest income. However, there is also the risk of default on a greater principal.

While traditional loans usually pay a spread borrowers also have to pay a fee for having the ability to use a facility - even if it is not used. This fee should cover the banks cost of holding liquid assets in case of emergency funding of large outflows. We can in most cases include this fee in the current drawn portion of the facility that is treated as a term loan. Any additional fees for drawn amounts can be added to the spread, β, in equation (14).

3.5 Loan-level net interest income with partial default and prepayment

Our model representation for loan-level net interest income has so far focused on a realized path of events that can happen up until maturity T. For a particular realization of stressed market and economic rates the path is not known with certainty. The realized path depends on the event probabilities as well as randomness. As we have mentioned before, the event probabilities may also depend on extended previous path history which can make it hard to solve the model analytically - requiring path simulation. See Skoglund and Chen (2016) on the stress testing practices with behavioral risk models using conditional Markov iteration approaches and state transition simulation approaches to solve for expected stress paths in the models. In the case of path simulation models for stress testing our event model applies along each stress test realized path. The idiosyncratic averaging of realized net interest income and $\text{NII}_{P/L}$ attribution to events yields the expected stress situation. However, not all loan-level models focus on event simulation. Oftentimes the credit and prepayment effects are modeled as rates on outstanding balances. This can for example happen in pool-level models or loan-level models that satisfy the conditional Markov property with respect to the realized stress path. Due to their nature such models often focus on the competing risks of prepayment and outright default rather than the possible paths to default. Assume therefore that at payment time k there is known expected frequencies of marginal default rates, $D(k)$, loss rates, $L(k)$, and marginal prepayment rates, $P(k)$. Such rates may of course depend on economic paths and we can write the net interest income, for a loan with a constant unity balance, at time $t = 1$ as

$$(R_A - R_f) \equiv (r_f(T) + \beta) - L(1) - r_f(T)$$

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with no loss of accrued interest for the defaulted portion. For $t = 2$ we have

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) B_b(1) \\ & -L(2) + B_1(1)(f_A(1,T) + \beta) \\ & -r_f(T)\end{aligned}$$

with

$$B_b(1) = 1 - [D(1) + P(1)]$$

the behavioral balance that is no longer paying the contractual rate at $t = 2$ such that we reduce the loan portion paying the contractual rate, $r_f(T)+\beta$, with the time $t = 1$ incremental default and prepayment rates. The incremental default and prepayment portion,

$$B_1(1) = D(1) + P(1)$$

now pays the forward rate plus spread i.e., $f_A(1,T) + \beta$. Now, at $t = 3$, incremental defaults and prepayments can in principle come from both the original contractual portion as well as the re-originated portions at the forward rate. Assuming, for simplicity of exposition, that default rates, prepayment rates and loss rates do not depend on the rate paid by the loan portion, we have at $t = 3$

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) B_b(2) \\ & -L(3) + B_1(2)(f_A(1,T) + \beta) \\ & +B_2(2)(f_A(2,T) + \beta) \\ & -r_f(T)\end{aligned}$$

where

$$B_b(2) = B_b(1) - B_2(2) B_b(1)$$

with

$$B_2(2) = D(2) + P(2)$$

representing the contractual portion furter reduced behavioral balance from the $t = 2$ default and prepayment rates. Then,

$$B_1(2) = B_1(1) - B_2(2) B_1(1)$$

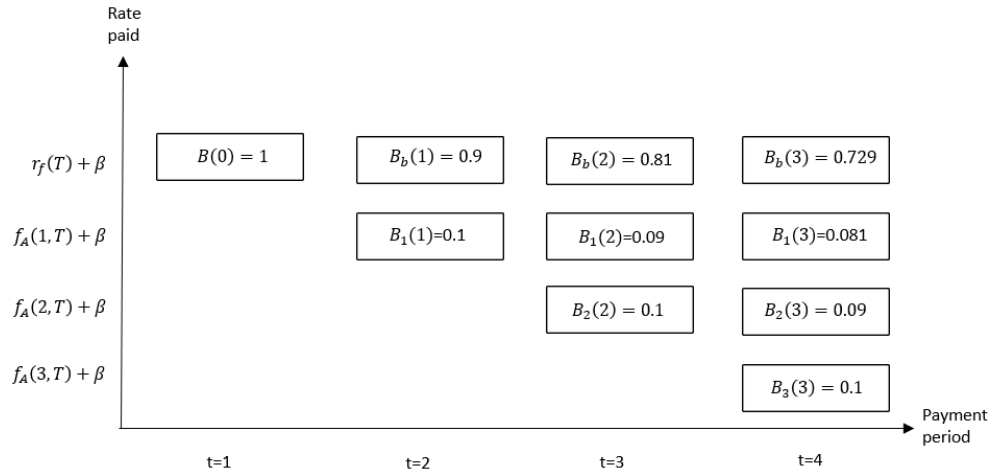
is the remaining balance of the re-originated portion at $t = 1$ still paying the foward rate, $f_A(1,T)$, at $t = 3$. This balance is reduced from $t = 2$ due to defaults and prepayments in the re-originated portion. This reduction is coming from the $t = 2$ default and prepayment rates. There is now also an amount of $B_2(2)$ balance paying the forward rate, $f_A(2,T)$, plus spread.

At time $t = 4$

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) B_b(3) \\ & -L(4) + B_1(3)(f_A(1,T) + \beta) \\ & +B_2(3)(f_A(2,T) + \beta) + B_3(3)(f_A(3,T) + \beta) \\ & -r_f(T)\end{aligned}$$

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Figure 3 Example balance movements between original contractual rate and forward rates in the partial default and prepayment rate model



with

$$B_b\left(3\right)=B_b\left(2\right)-B_3(3)B_b\left(2\right)$$

the previous behavioral principal still paying the contractual rate with

$$B_3(3)=D\left(3\right)+P\left(3\right)$$

and,

$$B_1\left(3\right)=B_1\left(2\right)-B_3(3)B_1\left(2\right)$$

the further reduced portion still paying the forward rate, $f_A\left(1,T\right)$. The reduction of the portion paying the forward rate, $f_A\left(2,T\right)$, is described by

$$B_2\left(3\right)=B_2(2)-B_3(3)B_2(2).$$

It is hence reduced by the incremental default and prepayment rates at $t=3$. The previous periods re-originations paying the forward rate, $f_A\left(3,T\right)$, are added with balance equal to the incremental default and prepayment rates, $B_3(3)$. From this example of the loan net interest income with partial default and prepayment rates it should be clear that the following net interest income periods for $t=5,...,T$ will move more and more portions of the loan to the prevailing forward rates. These portions are coming from both the original contractual portion as well as the previously re-originated balances. The balance movements, highlighted in our example above, to different earnings rates are described in Figure 3 when the sum of marginal default and prepayment rates are 10% each period.

An implementation of the partial default and prepayment model in the same spirit as the event model is not as straightforward as it may seem from our example. Especially, if we take into account that the default, loss and prepayment rates for the re-originated portions can differ from the original contractual portions. This is reasonable because they are paying a different rate and this rate should impact both the default rate as well as the prepayment rate. For example, when scenario forward rates are lower than the funding rate the prepayment incentive is reduced for the already re-originated portions. The implementation of

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the partial default and prepayment rate model, consistent with the event path framework for net interest income, hence requires that the modeler keeps track of the applicable default, loss and prepayment rates as well as the balance change over time for the corresponding portions paying the contractual and respective forward rates.

Instead of using the full partial default and prepayment rate model we will approximate the net interest income process with a partial model that assumes that incremental default and prepayment rates are only coming from the original contractual portion, while still on the full outstanding contractual balance. Since, incremental default and prepayment portions are only coming from the contractual portion we can safely assume that a single default rate and prepayment rate path is sufficient. This assumption gives the net interest income at a payment time t as,

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) \times \left(B_A(t-1) - \sum_{k=2}^{t} \left[D(k-1) + P(k-1) \right] B_A(k-2) \right) \\ & -L(t)\, B_A(t-1) + \sum_{k=2}^{t} \left[D(k-1) + P(k-1) \right] (f_A(k-1, T) + \beta)\, B_A(k-2) \\ & -r_f(T)\, B_A(t-1)\end{aligned} \tag{15}$$

with no loss of the accrued interest in case of default. Equation (15) can be motivated with the fact that for attribution purposes we may still want to attribute to the original loan. Hence, all defaults and prepayments happen to the contractual part. The partial default and prepayment rate model in equation (15) replaces any potential forward rate interest rate risk impact from re-originated portions, that could partially re-price again at the newer forward rates, with a re-pricing (of that part) of the original contractual rate portions at the forward rate. It moves more contractual portions to the forward rates and does not remove portions priced at old forward rates. Using the example in Figure 3 we have at $t = 4$ a contractual balance of 0.7 and a common balance of 0.1 for the re-origination parts. The difference between the attribution of balance to the contractual and re-origination portions will amplify over time. However, this effect is counteracted by reduced contractual principal and that when a loan is closer to maturity the likelihood of default and prepayment is usually smaller. It is also the case that stress test horizons are rarely longer than a few years. Still, allowing the default and prepayments to come from the re-originated portions, already paying a forward rate, results in different prepayment, interest rate and credit risk behavior. The modeler has to decide if equation (15) is a good enough approximation to the loan net interest income process or if also the re-originations should be modeled as having default and prepayment risk as well - with their own default, loss and prepayment rates. In this paper we focus on the approximation in equation (15) as a simple expression for the net interest income dynamics with behavioral and interest rate risk.

We have here assumed in equation (15) that there is no loss of the accrued interest $(r_f(T) + \beta) B_A(t)$ in case of default such that we only reduce the time k balance that earns the rate $r_f(T) + \beta$ at the next payment time, $k + 1$. This is reflected in the term

$$B_A(t-1) - \sum_{k=2}^{t} \left[D(k-1) + P(k-1) \right] B_A(k-2).$$

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This term can be adjusted for time k loss of accrued interest rate as follows

$$B_A(t-1) - \sum_{k=1}^{t} D(k) B_A(k-1) + \sum_{k=2}^{t} P(k-1) B_A(k-2)$$

which would reduce the time k interest earnings with the defaulted portion that does not pay it. Prepayments are assumed to include the accrued interest in the prepayment amount. Since the default rate term, $D(k)$, and the prepayment term, $P(k)$, are marginal (i.e., incremental rates) we sum their cumulative impact up until time t on the reduced balance earning. Essentially, the balance that earns the spread β over the matched maturity funds rate is decreasing due to credit defaults and early prepayments. It is replaced by earnings at the forward rate plus spread, $f_A(k,T) + \beta$. Hence, default and prepayments now act as haircuts on the balance $B_A(t)$ that still receives the contractual net interest income, $(r_f(T) + \beta) - r_f(T)$. When marginal default or prepayment rates are positive at time k part of the contractual net interest income of the next period moves to forward rate interest income i.e., $(f_A(k,T) + \beta) - r_f(T)$. Loss rates at time t, $L(t)$, are booked as negative outflows at the time the defaults happen. As before, when credit losses happen the synthetic liability still has to be paid.

As a simple numerical example of equation (15) consider the funding rate, $r_f(T)$, at 2%, the spread β at 1%, future forward rates, default and prepayment rates at 1% with loss given default rates of 20%. Using these numerical values and an assumed original balance of 100 units of currency with contractual principal payment of 10 units of currency at each payment time we obtain the reduction in the net interest income stream (compared to the contractual net interest income) for $t = 1$ and $t = 2$ as 0.2, and, for $t = 3$ as 0.198. At $t = 1$ there is only contribution from the loss term, $L(1) B_A(0)$. At $t = 2$ the loss term contribution, $L(2) B_A(1)$, is 0.18 while the forward rate contribution is 0.02. At $t = 3$ the loss term contribution, $L(3) B_A(2)$, is 0.16 while the total forward rate contribution is 0.038. If we assume also loss of time k accrued interest it reduces the net interest income stream further by 0.03, 0.027 and 0.024 respectively at $t = 1, 2, 3$.

As for the event model of net interest income the partial model has rundown of principal due to contractual principal payments only. It might be tempting to simply keep the current balances constant or scale the current balances upwards to a target balance over time to introduce new business. However, a large single loan may not behave the same way as a set of smaller loans. More importantly, simply adjusting balances of existing fixed rate loans ignores the fact that 100% of the new business balances should be priced at the scenario forward rate at the entering horizon.

Similarly to equation (15) we can write the partial model of net interest income at time t for a floating rate loan as

$$\begin{aligned} (R_A - R_f) &\equiv (f_A(t,t+1) + \beta) B_A(t-1) - L(t) B_A(t-1) - f_A(t,t+1) B_A(t-1) \\ &\equiv \beta B_A(t-1) - L(t) B_A(t-1). \end{aligned} \tag{16}$$

This is because the interest rate risk prepayment effect, $P(k)$, disappears as prepaid amounts are funded at the short-term forward rate, $f_A(t,t+1)$. The only effect is the credit loss rate, $L(t)$, similar to equation (13) although the loss rate is here a default probability weighted loss

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given default rate. Without loss of generality we can consider a floating rate loan that resets less frequent than the payment dates $t = 1, ..., T$ as a fixed rate loan up until the next reset time. Such a loan still has interest rate risk but the risk is reduced compared to a long-term fixed rate loan.

Both equation (15) and equation (16) show why the loan spread β must be priced to cover the behavioral risks. If the loan spread is not sufficient there is a risk of negative loan net interest income on a portfolio basis. This result is consistent with the credit risk spread framework. In practice the loan spread of non-traded risks like loans should cover expected losses, capital costs and embedded optionality. See Skoglund and Chen (2015) chapter 7.

We can of course represent the additional drawn portions of facilities for the partial models similar to equation (14). The difference is that with the partial model the facility drawn amount is not state dependent but it can of course be made dependent upon the severity of the economic stress scenario.

4 Application of stressed net interest income

In this section we illustrate with numerical examples the calculation and attribution of stressed net interest income using our matched maturity model. Our first model example uses simple default and prepayment rates. The assumption is that there is a model that can generate such default and prepayment rates conditional on economic paths. For example, a Cox proportional hazard model has a hazard rate for a particular loan or loan-segment i of the form

$$h^i(t) = \overline{h}(t) \exp(\boldsymbol{\beta}_i' \mathbf{Z}_t) \tag{17}$$

where $\boldsymbol{\beta}_i$ are the loan i specific factor loadings and the proportional term, $\overline{h}(t)$, introduces the non-stationarity baseline of the hazard rate over time - independent of the systematic factors, $\mathbf{Z}_t$. Such models are frequently used for default and prepayment rate models. In fact, any model that can be solved to generate default and prepayment frequencies conditional on a path for economic factors, $\{\mathbf{Z}_t\}_{t=1}^T$, fits within this framework. This includes the large class of state transition matrix models that can be solved using Markov iteration conditional on the economic factors, $\mathbf{Z}_t$, path. Examples include corporate multi-factor state transition models founded in the Merton (1974) framework of corporate default modeling as well as reduced form transition matrix models. See Skoglund and Chen (2016) for an overview of the application of credit models to the macroeconomic scenario analysis and stress testing.

Our second numerical example uses state transition models with state path simulation conditional on a path for economic factors, $\{\mathbf{Z}_t\}_{t=1}^T$, similar to the model in Figure 2. In practice, brute force event path simulation is usually reserved for models with significant past state history tracking. That is, models that cannot be easily solved using a Markov iteration conditional on the economic factors.

4.1 Default rates and prepayment rates

We consider the case of a fixed rate loan with 5 payment periods $t = 1, ..., 5$. To specify the contractual net interest income we need to specify the current funding curve. We assume the banks funding curve has percentage rates 1.1, 1.2, 1.3, 1.4, and, 1.5 respectively at the

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payment periods term, $t = 1, ..., 5$. Hence, the banks funding rate term-structure is increasing. The fixed rate loan spread β is 1 percent and the 100 units of currency balance remains fixed over the payment term. It is hence a bullet loan. Of course, since we assume the loan is always matched with a synthetic matched maturity funding the exact balance assumption is not important.

Now, at origination at time $t = 0$ this fixed rate loan has a matched maturity liability paying $r_f(T) = 1.5\%$. The contractual interest income is $r_f(T) + \beta$. Consider now a scenario shift of the banks funding rates at time $t = 0 + \epsilon$. Since the scenario shift of the curve happens at this time it affects any future rates set that will be paid at $t + 1$. We will use funding curve shifts at time $t = 0 + \epsilon$ to illustrate the interest rate effects. Equivalently, we could have specified a path for forward rates i.e., $f(t, T)$ at all future times t. However, our zero curve will also yield such a path for the forward rates calculated from the implied forward rates of the shifted zero curve.

To complete our bank balance sheet we also consider a Treasury that is funding with short-term rollover. Hence, the rollover is re-priced at funds forward rate $f(t, t + 1)$. This rate is obtained as the short-term implied forward rate of the funding curve. With contractual cash flows realized we have now locked in a spread of 1 unit of currency for each payment time t on the loan side. When the banks funding rates remain constant the Treasury net interest income, calculated using the forward rate, $f(t, t+1)$, is realized as 0.40, 0.20, 0, -0.20, -0.40 at the corresponding payment times, $t = 1, ..., 5$. This realized Treasury net interest income stream is due to the increasing forward rates, $f(t, t+1)$, which in turn is due to the increasing zero rates. It is well-known that when the zero curve is upward sloping the forward rates are greater than the zero rates. Similarly, if the zero curve is downward sloping the forward rates are less than the zero rates. However, with no shift of the current zero rate funding curve, the cumulative net interest income contribution from Treasury is 0.

If we now let the funding curve shift up and down in parallel respectively by 50 basis points the Treasury cumulative net interest income stream now realizes as -2 and $+2$ respectively. Clearly, the Treasury earns from lower forward rates and lower forward rates are realized from lower future funding zero rates. With Treasury re-financing interest rate risk it is natural to hedge part of that risk with market instruments. Assume therefore the Treasury enters into a swap agreement where it receives the floating leg on a notional of 100 units. Since the swap is priced off the market rates and not the banks fund rates we have to assume a zero rate term-structure of market rates. We let the current market rates be 10 basis points lower than the banks funding rates at all terms t with the difference attributed to the bank funding spread. Solving for the fair swap rate at $t = 0$ with these market rates we obtain the fair swap rate as approximately 1.394%. Applying the same up and down 50 basis points parallel jumps on both the funding rates and the market rates we obtain the Treasury cumulative net interest income after the swap P/L effects as $+0.364$ and -0.364 respectively. In practice there is basis risk of course – since the funding curve and market curve are not the same.

Since our main focus is behavioral risk on the loan side consider now a marginal prepayment rate at 1%. In our partial default and prepayment rate model in equation (15) the prepayment rates generate additional re-priced balances at payment times $t = 2, ..., 5$ at the incremental

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prepayment rates. With a constant balance $B_A(t) = B_A$ we can simplify equation (15) as

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) \times \left(1 - \sum_{k=2}^{t} [D(k-1) + P(k-1)]\right) B_A - L(t) B_A \\ & + \sum_{k=2}^{t} [D(k-1) + P(k-1)] (f_A(k-1, T) + \beta) B_A - r_f(T) B_A.\end{aligned}$$

If we also replace the delayed impact of prepayment and credit risk re-origination at the forward rate with an approximation assumption of accrued interest paid in arrears at the time k forward rate instead we have

$$\begin{aligned}(R_A - R_f) \equiv\ & (r_f(T) + \beta) \times \left(1 - \sum_{k=1}^{t} [D(k) + P(k)]\right) B_A - L(t) B_A \\ & + \sum_{k=1}^{t} [D(k) + P(k)] (f_A(k, T) + \beta) B_A - r_f(T) B_A.\end{aligned} \tag{18}$$

such that equation (18) further highlights the earnings rate substitution of the rate $r_f(T) + \beta$ with earnings at the rate, $f_A(k, T) + \beta$. Essentially, equation (18) accelerates the P/L effect on the net interest income from the forward rate impact - moving the effect from time $t+1$ to time t. We can immediately notice that if today's funding curve term-structure is increasing the implied forward rates > today's rates so a net earning is generated as the prepaid loan part interest income is increasing. In our example the exact cumulative $\text{NII}_{P/L}$ attribution is calculated as $+0.025$. Bumping up the funding rates by 50 basis points we get a larger cumulative $\text{NII}_{P/L}$ attribution at $+0.1$. In contrast, the cumulative $\text{NII}_{P/L}$ attribution for a downward shift of 50 basis points is -0.05. Hence, increased prepayment rates at lower future interest rates is a loan net interest income P/L risk.

In our combined example of loan origination net interest income and Treasury interest income we have that when the funding rates increase the actual Treasury funds cost increase and hence the Treasury net interest income P/L effect is negative. The loan origination net interest income effect is instead positive. However, loan prepayment is more likely when interest rates go down. The loan net interest income effect of prepayment is then negative. The effect on the Treasury depends on actual liabilities but in our example with short-term rollover at the forward rate lower interest rates yield a positive effect on the Treasury net interest income (i.e., without the swap P/L hedge). Hence, the Treasury short-term actual funding position hedges the prepayment interest rate risk.

We now introduce also default risk. We let the marginal default rate at all times t be 0.1%. With a constant loss given default rate at unity this means that the default loss rate effect is $0.001 \times 100 = 0.1$ units of currency at each payment time t. As for prepayment, the re-financed part contribution is positive with the current upward sloping zero curve. There are hence two effects. The positive re-financing contribution and the negative loss effect contribution. The total cumulative $\text{NII}_{P/L}$ attribution is -0.4975. This is less than the cumulative default loss contribution of -0.5 due to the re-financing effect. With both prepayment and default risk we have the cumulative loan $\text{NII}_{P/L}$ attribution as -0.4725. This is summable in prepayment and default impact. Specifically, the cumulative $\text{NII}_{P/L}$ attribution for default loss only is -0.5,

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the cumulative $\text{NII}_{P/L}$ attribution for the default re-financing part only is $+0.0025$, and, the cumulative $\text{NII}_{P/L}$ prepayment impact only is $+0.025$ such that

$$-0.4725 = -5 + 0.0025 + 0.025$$

With a constant funding curve the Treasury cumulative net interest income contribution is 0 as before.

From our simple analysis of loan net interest income it is clear that the loan cumulative $\text{NII}_{P/L}$ takes a negative hit from

- default loss rates being higher
- default re-financing with lower interest rates
- prepayment increases with lower interest rates

Consider therefore shifting the funding curve down by 50 basis points and also increasing the marginal default and prepayment rates by double. Low interest rates and high prepayment and default rates are certainly plausible. However, in practice, the prepayment incentive depends not only on interest rates but also on the ability to prepay and re-finance which can compensate for lower interest rates when the economy is doing badly in general. Hence, this can prevent large prepayments when interest rates are low together with stressed default rates. Still, applying these scenarios we obtain the loan cumulative $\text{NII}_{P/L}$ as -1.11. Hence, contractual cumulative net interest income (of 5) less the realized net interest income. This large negative loan P/L effect can be compensated for with our short-term Treasury funding strategy which would generate a cumulative $\text{NII}_{P/L}$ of $+2$ in case of the 50 basis points downward shift of the funding curve. Hence, the short-term Treasury funding is here a hedge. This gives a total loan and Treasury $\text{NII}_{P/L}$ of $+0.89$. If the Treasury hedges interest rate risk with the swap the total $\text{NII}_{P/L}$ is instead -1.544. This shows both the interaction and the possible separation between the loan net interest income and the Treasury net interest income.

4.2 A stochastic simulation example

Our example model of stochastic simulation of state path uses state transitions which contain a current state, five delinquent states being delinquent 30 days (D30), 60 days (D60), 90 days (D90), 120 days (D120) and 150 days (D150). Once a loan is delinquent 150 days (D150) it can only move back to current, stay in delinquent 150 days (D150) state, prepay or default. That is, there is no migration back to the other delinquent states. This gives the state transition matrix in Table 1.

In our state transition matrix model the transition function, $\theta_{lr}(t)$ can depend on the economic variables, $\mathbf{Z}(t)$, idiosyncratic variables, $\mathbf{X}_i$, and also on state transition history through past state indicator variables $\mathbf{I}_{it}$. This gives

$$\theta_{lr}(t) = g_{lr}\left(\mathbf{Z}_t, \mathbf{X}_i, \mathbf{I}_{it}, \boldsymbol{\beta}^{lr}\right).$$

with $\boldsymbol{\beta}^{lr}$ the parameter vector.

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Table 1 State transition matrix

	Current	D30	D60	D90	D120	D150	Prepay	Default
Current	θ_{11}	θ_{12}	0	0	0	0	θ_{17}	0
D30	θ_{21}	θ_{22}	θ_{23}	0	0	0	θ_{27}	0
D60	θ_{31}	θ_{32}	θ_{33}	θ_{34}	0	0	θ_{37}	0
D90	θ_{41}	θ_{42}	θ_{43}	θ_{44}	θ_{45}	0	θ_{47}	0
D120	θ_{51}	θ_{52}	θ_{53}	θ_{54}	θ_{55}	θ_{56}	θ_{57}	0
D150	θ_{61}	0	0	0	0	θ_{66}	θ_{67}	θ_{68}

To use the state transition matrix in Table 1 for stress testing we need to specify all the transition intensity models, $\theta_{lr}(t)$. We use individual proportional hazard models with a constant unity baseline (see equation (17)) for each of the transition intensities in the state transition matrix in Table 1. For simplicity, consider a single economic factor, Z_t, and the specific state path tracking delinquency indicators that follows:

- The months since last delinquent, ranging from $1, ..., 24$ months.
- Delinquency state occupancy indicator functions for states of delinquent 60 days (D60), 90 days (D90), 120 days (D120) and 150 days (D150).

Our specific state transition model is now

$$\theta_{lr}(t) = \exp\left(\alpha_i^{lr} + Z_t\beta^{lr} + \mathbf{I}_{it}\boldsymbol{\lambda}_i^{lr}\right) \tag{19}$$

where α_i^{lr} is a static contribution. The impact of the macroeconomics is measured through the parameter β_i with a single economic factor, Z_t. Here Z_t takes negative values in economic stress and positive values in economic upturns. The specific delinquency variables are captured in the indicator variables, $\mathbf{I}_{it}$, with impact parameters, $\boldsymbol{\lambda}_i^{lr}$. We set the parameters in the model such that past delinquency increases state transition probabilities to worse states. Similarly, when setting β_i we use that fact that when $\beta_i > 0$ in the model (19) negative values for Z_t reduces the transition intensity. Consequently, when $\beta_i < 0$ in the model (19) negative values for Z_t increases the transition intensity. As a general principle for the economic factor model parameters we therefore set $\beta_i > 0$ for transition intensities from worse states to better states (more delinquent to less delinquent) and $\beta_i < 0$ for transition intensities from better states to worse states (less delinquent to more delinquent or default). We also assume here that delinquent state prepayment is reduced in bad economic states and hence $\beta_i > 0$ for all the delinquent state transition intensities to prepayment (Krainer and Laderman (2011)). In this model the impact of the delinquency indicator functions is specified such that if it is not the first delinquency spell it increases the transition intensity back to less delinquent states. If the loan has been delinquent in a certain delinquency state before it also reduces the prepayment probability and the transition intensity to worse states (more delinquent or default). See appendix A for the exact model specifications. In general, a loan that is current at the start of the analysis has very little default risk. For example, from appendix A, the transition from current to delinquent 30 days (D30) state is

$$\theta_{12} = \exp\left(-4.5 - 0.5Z_t - 0.15I_{it}\right) \tag{20}$$

where I_{it} tracks the months since last delinquent, ranging from $1, ..., 24$ months. A loan that has never been delinquent is here treated the same as a loan that was delinquent at

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least 24 months ago. When Z_t is neutral ($Z_t = 0$) and the loan was delinquent more than 24 months ago we obtain the base transition rate $\theta_{12} \approx 0.0003035$ from equation (20). In addition, a current loan has to pass through all the increasingly severe delinquent states to ultimately default. While the stress on Z_t can increase the delinquency probability and ultimately default probability (e.g., $\theta_{12} \approx 0.0013604$ when $Z_t = -3$ with no past delinquency spell and $\theta_{12} \approx 0.0497871$ when $Z_t = -3$ and the loan was delinquent the previous period) the default event is still remote for a current loan in this state transition matrix specification. In contrast, the prepayment probability from the current state is larger

$$\theta_{17} = \exp(-4.5) \approx 0.0111. \tag{21}$$

While the default event is remote we still expect a relatively large contribution from loan paths where delinquency spells happens which may not be realized in outright default. In delinquency spells the interest earnings on the loan accrue and may not be paid back if the delinquency situation does not improve. The synthetic liability still has to be paid. A loan that becomes delinquent and cures before the end of the stress test period has no impact on the cumulative net interest income as the loan must pay the accruals to become current again. There can also be partial cures.

In general the loan net interest income of a fixed rate loan (see equation (12)) in our state transition model depends on:

- Macroeconomic factors driving delinquency, prepayment and default together with the exact state path history. Hence, to stress the model we require a path for Z_t. On each simulation state path $\{Z_t\}_{t=1}^T$ remains at stressed values in our example. In practice, the stressed path can of course include a mean-reversion to a neutral baseline i.e., $Z_t = 0$. Regardless of the specific stressed path for Z_t the different idiosyncratic realizations along the given stress path are evaluated.

- Factors driving the loss given default rate which can also be influenced by macroeconomic variables. For example, a mortgage loss given default is closely connected to its loan to value at default which in turn depends on the property prices path, say $\{S_t\}_{t=1}^T$. However, at default, we will for simplicity assume there is a constant loss given default rate of unity in our example.

- The forward rates, $\{f_A(t,T)\}_{t=1}^T$, at the time of re-financing. We can here specify a complete forward rate path, $\{f_A(t,T)\}_{t=1}^T$, as part of our comprehensive stress scenario. Alternatively, we can specify a stressed zero rate term-structure, $r_f(t,T)$, at a certain time t and obtain the stressed forward rates as implied forward rates from that term-structure. We choose to specify the complete forward rates, $\{f_A(t,T)\}_{t=1}^T$, in our example. In practice the forward rates come into play both for the re-originations as well as driving the prepayment rates. In our transition model specification the delinquent prepayment rates depend on the macroeconomic factor, Z_t, only while the prepayment rate for a current loan is constant (see equation (21)). We will however introduce a simple prepayment incentive with burnout for comparison.

- The current funding rate, $r_f(T)$, and the spread, β. We assume the current funding rate is 1% and the spread is 30 basis points i.e., 0.003.

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The stressed path $\{Z_t\}_{t=1}^T$ for $t = 1, ..., 24$ periods is set to a common value of -3. The corresponding downward stress for $\{f_A(t,T)\}_{t=1}^T$ moves all future forward rates down from 1% to 0.05% or 0.01%. We also consider an upward shift of the future forward rates to 0.015%. Our model uses monthly simulation periods due to the monthly delinquency structure of the state transition matrix and assume the payments happen on monthly periods as well. This means $\Delta t = \frac{1}{12}$ for each payment period. The balance is assumed constant at $10,000$ units of currency. That is, $B_A(t) = 10,000$ during the stress period. The loan is a bullet loan such that the forward rate, $f_A(t,T)$, coincides with the matched maturity forward rate. This gives a contractual net interest income per period as

$$(R_A - R_f) \equiv (0.01 + 0.003 - 0.01) \times 10000 \times \frac{1}{12} = 2.5.$$

When there is delinquency interest earnings accrue until delinquency improves or is ultimately lost in default. There is no late fee.

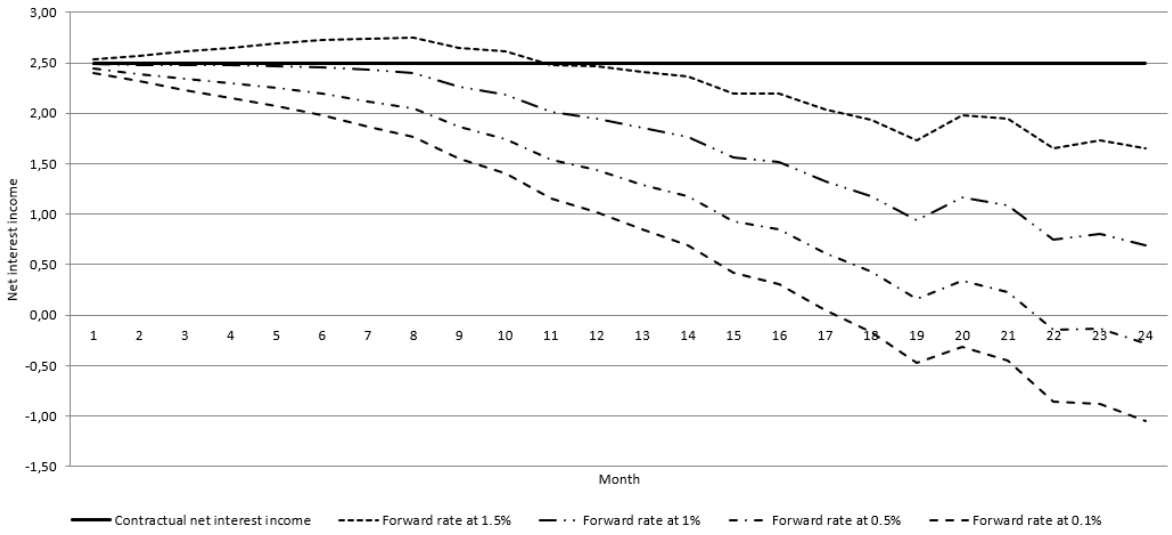
Consider now a premium quality fixed rate loan that is starting out as current and with no delinquency history[9]. Figure 4 displays the net interest income realized by state simulation using the state transition matrix in Table 1 and the net interest income process in equation (12) with both credit and interest rate risk. There are different scenarios for the forward rate corresponding to a constant future forward rate of 1.5%, 1%, 0.5% or 0.1%. The case when the forward rate is 1% keeps the forward rate path, $\{f_A(t,T)\}_{t=1}^T$, at the baseline case for $t = 1, ..., 24$ periods. This is interpreted as the stressed loan net interest income path with only credit default and delinquency risk. The net interest income results in Figure 4 are interpreted as the average stressed loan net interest income path, conditional on the stressed realizations of the macroeconomic factor and the forward rate, for a very large number of idiosyncratic state path simulations. Due to the remote risk of ultimately default for this loan the majority of credit risk losses along the stress path are attributed to delinquency spells that do not cure or only cure partially. Hence, with limited outright defaults the interest rate risk effect is coming largely from prepayment for this loan. This is illustrated with the impact of the different forward rate scenarios on the net interest income. A higher forward rate yields initial larger net interest income due to the repricing of prepaid loans. However, eventually there is a loss effect from default and delinquency spells that cannot be mitigated with increased forward rates. The lower forward rate scenarios means that prepaid loans are re-priced at lower rates while the synthetic liability interest payment remains fixed. The effect is increasing with decreasing forward rates. While the prepayment effect for delinquent loans depends on the macroeconomic scenario and the path behavior (see appendix A) the prepayment level for current loans is constant (see equation (21)). This means the majority of prepayment effects are fixed. Together with the flat scenarios for the forward rates this is resulting in a shift upwards/downwards compared to the net interest income with credit risk (default and delinquency) effect only. As for the default and prepayment rates example we have assumed the time t prepayment and default event accrual is paid in arrears at the forward rate, $f_A(t,T)$. This accelerates the interest rate effect from time $t+1$ to time t.

The constant prepayment rate for current loans assumed in Figure 4 yields an almost flat prepayment level. However, if we introduce a simple increased prepayment incentive into

[9]In our example, when a default or prepayment event happens along a simulated path, we reset the delinquency history to a fresh start as current again.

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Figure 4 Stressed net interest income



equation (21) as follows

$$\theta_{17} = \exp\left(-4.5 + \frac{1}{j}\right) \tag{22}$$

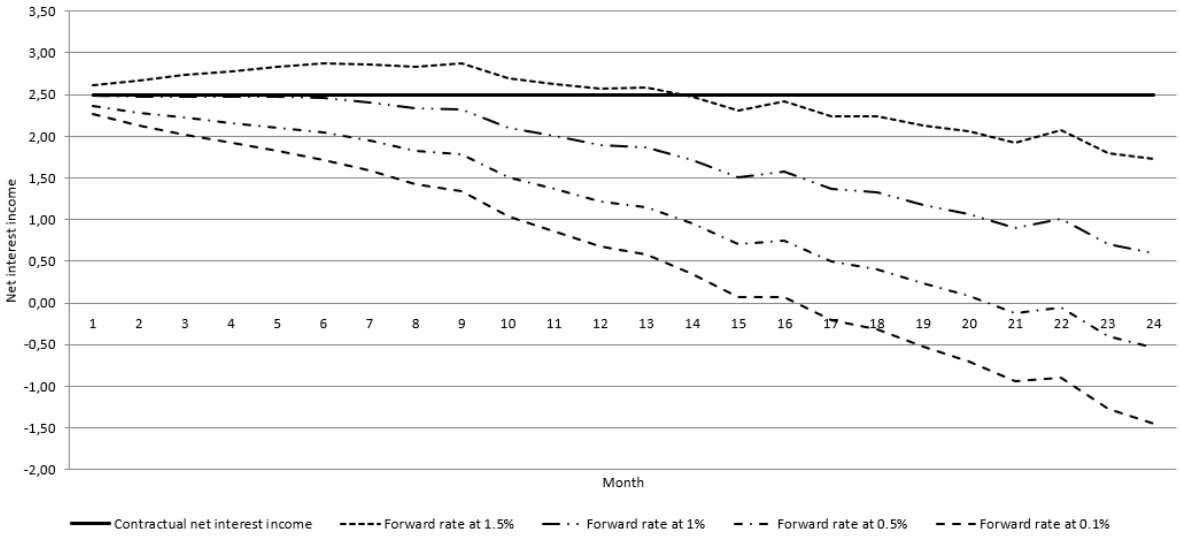
with j the count of the number of months in the projection we have that the prepayment rate is higher than in equation (21) at the start, capturing the increased incentive when lower forward rates are realizing. As the count of the number of months in the projection increase the prepayment rate lowers to capture a burnout effect. At the first projection month the prepayment rate for current loans is approximately 0.03019 in equation (22) and then decreases fast to similar levels of prepayment as in equation (21). Figure 5 displays the net interest income with credit risk and interest rate risk when we use the adjusted prepayment model for current loans with increased prepayment incentive and burnout. We consider all the forward rate scenarios as in Figure 4 recognizing that the prepayment incentive effect is more relevant for the downward forward rate scenarios. As expected for the downward forward rate scenarios there is slightly lower net interest income compared to Figure 4. Similarly, there is slightly larger net interest income compared to the increasing forward rate case in Figure 4. In general there can be a competing risk effect in the sense that an increased prepayment incentive can mitigate delinquency/default effects. The exact outcome on the net interest income projection depends on the relation between prepayment repricing interest rate effects and delinquency/default effects.

5 Summary and conclusions

The macroeconomic stress testing of net interest income is an important contributor to the stressed capital adequacy analysis process in Figure 1. Traditionally, the stress testing of credit losses and interest rate risk has been done in silos due to the historical legacy of asset and liability management systems doing the valuation of liabilities and the credit stress testing systems focusing on the behavioral risks. This can result in inconsistencies - especially from

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Figure 5 Stressed net interest income with increased prepayment incentive and burnout



behavioral risks that affect the interest income in the following stress period, and interest rate changes that influence the behavioral risks. Our loan level net interest income definition allows the separation of the cash flow valuation of liabilities with the evaluation of behavioral risks of the assets while taking into account the interactions. This is an important result because the complexity of joint behavioral and interest rate stress test model execution can be significantly reduced. In addition, our loan level net interest income definition allows the P/L attribution of risks down to granular level and risk type which can increase insight not only into where the main stress test contributors reside but also why.

We would like to end the paper with a few words about the practical implementation of the stressed net interest income framework.

- Firstly, it is clear that the changes to existing behavioral risk models is small. In fact, with our framework, cash flow generation can be avoided for the behavioral risks which closely resembles the stressed credit impairment process in IFRS 9 and CECL where outstanding balances are sufficient.

- Secondly, there is the question of defining the matched maturity funding rate for loans. For new originations this rate is well-known for banks with a funds transfer pricing practice. This is also true for most historical originations. If the matched maturity rate is not available it can in many cases be calculated exactly or approximated with access to the banks historical funds curve.

- Thirdly, we have the loan spread. The spread is defined from the difference of the current paying rate and the matched maturity funds rate. Knowledge about the banks own historical behavior over time for setting spreads to different loans and under different environments and behavioral risks may help understand if the re-origination spread needs to be modeled or can be kept constant.

- Fourthly, there is the re-origination at forward rates. This forward rate is in principle

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a matched maturity rate. It can be calculated from the scenario realized interest rate term-structure. It can also be approximated.

- Fifthly, there is the Treasury side that generates interest rate risk stressed cash flows for the synthetic assets versus the actual liabilities. It is important to understand that the synthetic assets are here simple certain contractual delivery instruments i.e., without behavioral risk. That is, the cash flow complexity is with the actual liabilities that always have to be valued. Moreover, only the aggregated portfolio P/L is of interest.

Appendix B provides further details on the step-by-step implementation of our integrated loan net interest income framework.

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References

1. Bank of England (2012), RAMSI: A Top-Down Stress Testing Model, Stability Paper No 17, September.

2. Bank of International Settlements (2018), Stress Testing Banks - A Comparative Analysis, Financial Stability Institute, FSI Insights on Policy Implementation No 12.

3. Dev, A and Rao, V (2006), Performance Measurements in Financial Institutions in an ERM Framework, Risk Books.

4. De Nederlandsche Bank (2017), A Top-Down Stress Testing Framework for the Dutch Banking Sector, Occasional Studies Volume 15-3.

5. Hall, A and Lundstedt, K. G (2005), The Competing Risks Framework for Mortgages: Modeling the Interaction of Prepayment and Default, The RMA Journal, September.

6. Jarrow, R and van Deventer, D (1998), The Arbitrage-Free Valuation and Hedging of Demand Deposits and Credit Card Loans. Journal of Banking and Finance 22, pp. 249–272.

7. Krainer, J and Laderman, E (2011), Prepayment and Delinquency in the Mortgage Crisis Period, Federal Reserve Bank of San Francisco, Working Paper 2011-25.

8. Levin, A (2001), Active-Passive Decomposition in Burnout Modeling, Journal of Fixed Income, Vol 10 (4), pp. 27–40.

9. Levin, A and Davidson, A (2005), Prepayment Risk- and Option-Adjusted Valuation of MBS - Opportunities for Arbitrage, Journal of Portfolio Management, 31 (4), pp. 1–20.

10. McConnell, J and Singh, M (1994), Rational Prepayments and the Valuation of Collateralized Mortgage Obligations, Journal of Finance, Vol 49 (3), pp. 891–921.

11. Merton, R. C (1974), On the Pricing of Corporate Debt: The Risk Structure of Interest Rates, Journal of Finance, Vol 29 (2), pp. 449-470.

12. Nyström, K and Skoglund, J (2006), A Credit Risk Model for Large Dimensional Portfolios with Application to Economic Capital, Journal of Banking & Finance, 30, pp. 2163-2197.

13. Nyström, K (2008). On Deposit Volumes and the Valuation of Non-Maturing Liabilities, Journal of Economic Dynamics and Control. 32, pp. 709-756.

14. Skoglund, J (2017), Loan Fair Value Approaches Revisited,

    Available at SSRN: https://ssrn.com/abstract=2902564.

15. Skoglund, J and Chen, W (2015), Financial Risk Management: Applications in Market, Credit, Asset and Liability Management and Firmwide Risk, Wiley Finance Series.

Electronic copy available at: https://ssrn.com/abstract=3547286

16. Skoglund, J and Chen, W (2016), The Application of Credit Risk Models to Macroeconomic Scenario Analysis and Stress Testing, Journal of Credit Risk, vol 12 (2), pp. 1-45.

17. Skoglund, J and Chen, W (2018), Stage Transfer Effect on Impairment Forecasts, Journal of Risk Management in Financial Institutions, vol 11 (3), pp. 244-256.

18. Stanton, R (1995), Rational Prepayment and the Valuation of Mortgage-Backed Securities, Review of Financial Studies, 8 (3), pp. 677–708.

19. Tilloca, D (2015), A Risk-Neutral Approach for the Evaluation of Commercial Loans, Available at SSRN: https://ssrn.com/abstract=2576245.

20. Tschirhart, J, O'Brien, J, Moise, M and Yang E, (2007), Bank Commercial Loan Fair Value Practices, Federal Reserve Board, Washington, 2007-29.

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A State transition models

For the state transition simulation models using the transition matrix in Table 1 we first let transitions from the current state be modeled as

$$\begin{aligned}\theta_{12} &= \exp\left(-4.5 - 0.5Z_t - 0.15I_{it}\right)\\ \theta_{17} &= \exp\left(-4.5\right)\end{aligned}$$

where I_{it} now tracks the months since last delinquent, ranging from $1, ..., 24$ months. A loan that has never been delinquent is here treated the same as a loan that was delinquent at least 24 months ago. The transition from current state back to current, θ_{11}, is set to unity before normalization. In our case we use a simple normalization such that we obtain a normalized θ_{lr}, $\widetilde{\theta}_{lr}$, with

$$\widetilde{\theta}_{lr} = \frac{\theta_{lr}}{\sum_{r=1}^{K}\theta_{lr}}.$$

Since we will set all transition which has the same from and to state to unity (before normalization) we do not need to specify them explicitly hereafter.

The delinquent 30 days (D30) row of the transition matrix in Table 1 has the transition model specifications

$$\begin{aligned}\theta_{21} &= \exp\left(-1.5 + 0.5Z_t + 0.17\overline{I}_{it}\right)\\ \theta_{23} &= \exp\left(-2 - 0.5Z_t - 0.09\overline{I}_{it}\right)\\ \theta_{27} &= \exp\left(-3.3 + 0.5Z_t - 0.11\overline{I}_{it}\right).\end{aligned}$$

Here, the delinquency indicator $\overline{I}_{it}$ is now unity if the loan has ever been delinquent 60 days (D60) and null otherwise. For the from state delinquent 60 days (D60) transition models in the transition matrix in Table 1 we have that

$$\begin{aligned}\theta_{31} &= \exp\left(-1.8 + 0.5Z_t + 0.17\widehat{I}_{it}\right)\\ \theta_{32} &= \exp\left(-1.5 + 0.5Z_t + 0.17\widehat{I}_{it}\right)\\ \theta_{34} &= \exp\left(-2 - 0.5Z_t - 0.09\widehat{I}_{it}\right)\\ \theta_{37} &= \exp\left(-3.3 + 0.5Z_t - 0.11\widehat{I}_{it}\right)\end{aligned}$$

where $\widehat{I}_{it}$ is now unity if the loan has ever been delinquent 90 days (D90) and null otherwise. For the transition intensities from state delinquent 90 days (D90) we have

$$\begin{aligned}\theta_{41} &= \exp\left(-1.8 + 0.5Z_t + 0.17\overrightarrow{I}_{it}\right)\\ \theta_{42} &= \exp\left(-1.5 + 0.5Z_t + 0.17\overrightarrow{I}_{it}\right)\\ \theta_{43} &= \exp\left(-1.3 + 0.5Z_t + 0.17\overrightarrow{I}_{it}\right)\\ \theta_{45} &= \exp\left(-2 - 0.5Z_t - 0.09\overrightarrow{I}_{it}\right)\\ \theta_{47} &= \exp\left(-3.3 + 0.5Z_t - 0.11\overrightarrow{I}_{it}\right)\end{aligned}$$

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with $\overrightarrow{I}_{it}$ unity if the loan has ever been delinquent 120 days (D120) and null otherwise. We also have the transition intensities from state delinquent 120 days (D120) as

$$\begin{aligned}
\theta_{51} &= \exp\left(-1.8 + 0.5Z_t + 0.17\overleftarrow{I}_{it}\right) \\
\theta_{52} &= \exp\left(-1.5 + 0.5Z_t + 0.17\overleftarrow{I}_{it}\right) \\
\theta_{53} &= \exp\left(-1.3 + 0.5Z_t + 0.17\overleftarrow{I}_{it}\right) \\
\theta_{54} &= \exp\left(-1.2 + 0.5Z_t + 0.17\overleftarrow{I}_{it}\right) \\
\theta_{56} &= \exp\left(-2 - 0.5Z_t - 0.09\overleftarrow{I}_{it}\right) \\
\theta_{57} &= \exp\left(-3.3 + 0.5Z_t - 0.11\overleftarrow{I}_{it}\right)
\end{aligned}$$

where now, $\overleftarrow{I}_{it}$, is unity if the loan has ever been delinquent 150 days (D150) and null otherwise. The final transition models we need to consider are the transition models from state delinquent 150 days to current, prepayment or default. They are

$$\begin{aligned}
\theta_{61} &= \exp\left(-2 + 0.5Z_t + 0.17\overleftarrow{I}_{it}\right) \\
\theta_{67} &= \exp\left(-3.3 + 0.5Z_t - 0.11\overleftarrow{I}_{it}\right) \\
\theta_{68} &= \exp\left(-1.2 - 0.5Z_t - 0.09\overleftarrow{I}_{it}\right)
\end{aligned}$$

which again uses the indicator function $\overleftarrow{I}_{it}$. This completes the specification of the transition intensity models for the state transition matrix in Table 1.

B Step-by-step implementation

The implementation of the loan net interest income framework requires relatively small adjustments to existing model practices which is an advantage. Starting with the approximate partial default and prepayment rate model of the net interest income process in equation (15) for fixed rate loans, we have the following step-by-step implementation scheme.

1. For every loan or loan segment in the current book obtain the matched maturity funding rates and current spread.

2. For all the horizons on the scenario path, iterate on the following steps:

   (a) Start at the current principal balance.

   (b) Calculate marginal prepayment rates, default rates and loss given default rates using the scenario point in time behavioral models.

   (c) Calculate the incremental re-origination balances (from marginal prepayment and default rates) forward rate from the scenario yield curve term-structure. Potentially, consider also a spread change for the re-originations. Then add the effect of previous periods re-origination of balances due to default and prepayment.

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(d) Book the loss given default rate as a negative contribution.

(e) Write down the contractual balance that receives the original interest earnings by the cumulative prepayment and default rates.

(f) Add the calculated components above in the net interest income equation (15).

(g) Apply a discount term if the target is discounted net interest income.

3. Sum up the net interest income over the scenario horizons.

For the current book stress testing the above steps are sufficient. For a dynamic balance projection with target balance for certain segments, at each scenario horizon, apply the following additional steps:

4. Sum up the current balance at each horizon and find the gap against the target balance. This difference is only due to contractual principal payments and can therefore be calculated before the current book run in most cases.

5. Originate additional positions at the matched maturity rate plus the assumed spread and follow the steps 2a-2g at the start horizon of the new position. For target future positions the starting matched maturity rate is coming from the scenario yield curve term-structure.

6. Sum up as in step 3 the net interest income for the new target positions.

The corresponding implementation scheme for floating rate loans is much simpler. The net interest income process for floating rate loans in equation (16) only requires the current principal balance, the spread, and, the loss given default rate.

With some modifications the above logical steps apply to the loan net interest income event simulation models as well in equation (12) for the fixed rate loans. Specifically, we need to take into account the fact that a net interest income is now realized on the idiosyncratic state path rather than having available scenario rates of default and prepayment from the models. Moreover, there is an averaging process of all the idiosyncratic state paths. For example, in step 2b above we instead observe the state realized. A default or prepayment event on the path requires the calculation of the forward rate in step 2c. However, the full current principal balance is re-originated at the forward rate. Step 2d is the same in case of a default event. We then keep track of the net interest income equation realized until the next event happens. There can also be model state path history tracking adjustments made at an event as in our application example. In step 3 we both average the realized idiosyncratic net interest income paths as well as sum up over the scenario horizons. As for the partial model floating rate loans are much simpler since defaulted amounts are already re-priced at the forward rate. Hence, there is no interest rate risk - only loss given default risk and, potentially, spread risk of the re-origination.

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